Sebastian Tamayo

Senior Software Engineer

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Results-driven Senior Web Developer with over a decade of experience architecting, developing, and delivering scalable, high-performance web applications. Specializes in crafting intuitive, responsive user interfaces using React.js and building resilient backend systems with Python Django, Django REST Framework, Flask, FastAPI and cloud-native architectures.

Brings full-lifecycle expertise—from requirements gathering and system design to CI/CD implementation, automated testing, and long-term support. Thrives in Agile, cross-functional environments, fostering collaboration, rapid iteration, and strong alignment with business and product goals.

Recognized for writing clean, maintainable, and well-structured code, mentoring junior engineers, and championing SOLID principles, clean architecture, and Test-Driven Development (TDD). Adept at leading technical discussions, applying proven design patterns, and optimizing applications for performance, scalability, and reliability.

Committed to continuous learning, with a strong focus on modern web technologies, cloud infrastructure, and DevOps automation. Passionate about building user-centric software that solves real-world problems and delivers measurable business impact.

PROFESSIONAL EXPERIENCE

UST

Senior Software Engineer

June 2022 – June 2025 (3 years)

- Collaborated in a cross-functional **Agile** (**Scrum**) team, mentoring junior developers and conducting detailed code reviews emphasizing **SOLID** principles, design patterns, and clean architecture to uphold code quality and maintainability.
- Led the modernization of a high-traffic customer-facing platform using **React.js**, significantly enhancing accessibility, responsiveness, and client-side rendering. Integrated live auction streaming via **Azure Media Services** and connected seamlessly with the **Shopify Storefront API**, resulting in increased user engagement and conversions.
- Designed and maintained scalable backend services with Python frameworks including **Django**, **Flask**, and **FastAPI**, utilizing **PostgreSQL** and MongoDB for relational and **NoSQL** storage. Applied query optimization, schema tuning, and advanced indexing strategies, reducing data retrieval latency by over 20%.
- Architected a resilient microservices ecosystem on **Azure Kubernetes Service** (AKS), leveraging **Azure Service Bus** for asynchronous communication between distributed services, improving system modularity, scalability, and fault tolerance.
- Spearheaded the development of an AI-enabled chatbot using Azure Cognitive Services, Azure Functions, and Cosmos DB, delivering intelligent, real-time interactions and significantly enhancing customer support efficiency.
- Enhanced frontend and backend performance through React component optimization (lazy loading, memoization, suspense boundaries) and streamlined business logic, achieving a 30% improvement in load speed and user satisfaction.
- Implemented and maintained robust CI/CD pipelines using **Azure DevOps**, integrating automated testing, linting, and deployment workflows. Improved release velocity by 22% while maintaining quality through automated test suites built with **PyTest** (**Django**, **Flask**, **FastAPI** services), **Jest** (frontend), and **Cypress** (end-to-end).

Tech Stack: React.js, TypeScript, Redux, SCSS, Python, Django, Django REST Framework, Flask, FastAPI, PostgreSQL, MongoDB, Azure Kubernetes Service (AKS), Azure Service Bus, Azure Functions, Azure Cognitive Services, Azure Media Services, Cosmos DB, Shopify Storefront API, Docker, Kubernetes, Azure DevOps, PyTest, Jest, Cypress, Git, Jira, Postman

Gorilla Logic

Full Stack Engineer

February 2018 – May 2022 (4 years 4 months)

- Designed and developed responsive, mobile-first user interfaces using **React**.js, **TypeScript**, and **SCSS**, resulting in a 40% increase in mobile user retention and a 30% improvement in overall user engagement.
- Built scalable, modular frontend components for enterprise-grade dashboards, leveraging **Redux** for efficient state management and ensuring maintainability across large, cross-functional applications.
- Engineered secure, high-performance backend services with Python frameworks including **Django**, **Flask**, and **FastAPI**, applying microservices principles and integrating Apache Kafka to support real-time, event-driven communication in distributed environments.
- Automated infrastructure provisioning with **AWS CloudFormation** and deployed containerized services via Docker on **AWS ECS**, reducing deployment time by 50% and improving release velocity.
- Designed and implemented real-time analytics and sentiment analysis pipelines using **AWS Lambda**, **S3**, and **DynamoDB**, leading to a 35% improvement in AI model accuracy and data insight generation.
- Tuned backend performance through advanced **PostgreSQL** optimization techniques, including indexing strategies, query refactoring, and schema design, resulting in a 15% reduction in query latency.
- Managed distributed microservices architecture on Kubernetes, enabling high availability, fault tolerance, and seamless zero-downtime deployments across business-critical systems.
- Embraced **Test-Driven Development** (**TDD**) using **PyTest** and Jest, and implemented comprehensive observability practices with **AWS CloudWatch**, **Elastic APM**, and **Prometheus**/**Grafana** to support real-time diagnostics and proactive system monitoring.

Tech Stack: React.js, TypeScript, Redux, SCSS, Python, Django, Django REST Framework, Flask, FastAPI, PostgreSQL, Apache Kafka, REST API, AWS (Lambda, ECS, S3, CloudFormation, DynamoDB, CloudWatch), Docker, Kubernetes, Elastic APM, Prometheus, Grafana, PyTest, Jest, Git, Jira, Postman

Sophos Solutions

5 years 6 months

Senior Web Developer

September 2015 – February 2018 (2 years 6 months)

- Contributed to the development of a high-performance Single Page Application (SPA) for a financial services platform using **React**.js, **Redux**, and React Hooks, leading to a 30% increase in user engagement and enhanced application responsiveness.
- Refactored legacy PHP and Laravel systems into modular frontend components and transitioned backend services with Python frameworks including **Django**, **Flask**, and **FastAPI**, reducing server response times by 40% and improving scalability and maintainability.
- Applied advanced frontend performance optimizations such as code splitting, lazy loading, tree shaking, and image compression, decreasing initial load times by 35% and improving **SEO** with structured metadata and schema markup.
- Delivered mobile-first, AMP-compliant user interfaces with React, ensuring **WCAG** accessibility compliance and increasing mobile traffic by 25% across various platforms.
- Built comprehensive test coverage using **Jest, React Testing Library** and **Cypress**, integrated into Git-based **CI/CD** pipelines, reducing production defects by 75% and significantly enhancing release stability.
- Partnered with UX/UI designers using **Adobe XD** to revamp enterprise dashboards, improving navigation speed by 50% and boosting team productivity by 25% through cleaner design and improved usability.

Tech Stack:

React.js, TypeScript, Redux, React Hooks, SCSS, Python, Django, Django REST Framework, Flask, FastAPI, PostgreSQL, REST API, AMP, Jest, React Testing Library, Cypress, Git, GitHub Actions, Adobe XD, Jira, Postman

Web Developer

September 2012 – February 2015 (2 years 6 months)

- Developed responsive, accessible, and standards-compliant web applications using **React.js, TypeScript, HTML5**, and **CSS3**, following modern usability and **WCAG** accessibility guidelines.
- Engineered modular, reusable **React** components using functional programming paradigms, hooks (e.g., useEffect, useContext), and context-based state management to ensure scalable frontend architecture.
- Built robust and scalable RESTful APIs using Python frameworks such as **FastAPI** and **Django**, implementing asynchronous programming, middleware layers, and clean routing to support high-throughput enterprise systems.
- Streamlined build and deployment workflows leveraging **Webpack**, **Vite**, and Python packaging tools, optimizing environment-specific configurations, asset bundling, and production readiness to accelerate release cycles.
- Collaborated closely with **DevOps** and backend teams to integrate caching strategies using **Redis**, and to optimize CI/CD pipelines via **GitHub** Actions and **Azure DevOps**, ensuring reliable and automated deployments.
- Designed and optimized complex **PostgreSQL** and **MySQL** queries, including stored procedures and indexing strategies, to power analytical dashboards with improved performance and scalability.
- Integrated Google Analytics and custom telemetry solutions to monitor user behavior and system metrics, driving UX improvements that resulted in increased conversion rates.
- Achieved a 35% reduction in page load time through frontend optimization techniques such as lazy loading, tree shaking, image compression, and minification, significantly enhancing SEO rankings and user experience.

Tech Stack: React.js, TypeScript, Python (FastAPI, Django), REST API, PostgreSQL, MySQL, Redis, Azure DevOps, GitHub Actions, Webpack, Vite, Jest, React Testing Library, Google Analytics, HTML5, CSS3, SEO, WCAG, Telemetry

EDUCATION

Universidad Católica de Pereira 04.2005 – 09.2012

Bachelor of Computer Science

SKILLS

Languages:	C++, JS(ES6), TS, Python, C#, Rust, PHP, Java, HTML5
Frameworks and Libraries:	Django, Flask, Fast API, .Net Core, .Net MVC, React.js, Next.js, Gatesby, Angular, Node.js, Express, React Native, Redux, RxJS, REST API, GraphQL, Ably, Laravel, Tailwind CSS, RabbitMQ, Kafka, TypeORM, Shopify, WordPress, Entity Framework, JPA, Java Spring Boot
DBMS	MySQL, MS SQL, PostgreSQL, MongoDB, AWS DynamoDB, AWS RDS, Azure SQL Database
IDEs and Tools:	Visual Studio Code, Jenkins, Cypress, Jest, Jasmine, Terraform, Docker, Kubernetes, Redux DevTools, Babel, Webpack, Storybook, Azure DevOps, Git Bash, PowerShell
Cloud Services:	AWS Cloud Services, Microsoft Azure, GCP
Other skills:	Problem Solving, Full Stack Development, Critical Thinking, Leadership, Algorithm Design, Team-Oriented, SOLID, Agile (Scrum), Microservices, Test-Driven Development (TDD), Diagnosing, Business Analysts